

Early Cutaneous Manifestations of Lyme Disease

Erythema Migrans

Clinical Features

Erythema migrans (EM) is the hallmark skin lesion of early localized Lyme disease. In early case reports, EM was observed in 50% to 80% of patients with Lyme disease. More recent studies report lower incidence, likely due to increased awareness and early recognition of tick bites and symptoms.

A definite history of tick bite at the site of the lesion is reported in only a small proportion of patients. EM lesions typically develop 3 to 30 days after a tick bite, with a median onset of 7 days. The lesion is believed to result from direct invasion by *Borrelia burgdorferi*, supported by successful cultivation of the spirochete from EM lesions.

EM can occur anywhere on the body but shows site predilection based on age and tick attachment preferences: - In adults: most commonly appears on the lower extremities, groin, and axillary regions. - In children: frequently seen on the face, sometimes presenting with a "slapped cheek" appearance.

No clear sex predilection has been noted in the United States, although European studies report higher rates in women.

The lesion has a characteristic, pathognomonic appearance: - Begins as an expanding area of erythema centered on the tick bite site. - The border is typically confluent and well-defined against normal skin, while the transition from the central to peripheral zone is less distinct. - Often described as round, but more commonly oval, with its long axis aligned parallel to Langer's lines (lines of least skin tension). - As the lesion expands, its shape may become distorted. - Central clearing frequently occurs after several weeks, resulting in

an annular (ring-like) appearance with a red, expanding outer edge and a pale or normal-appearing center. - In very large lesions, only a segment of the erythematous border may be visible.

The classic "bull's-eye" morphology—central clearing with an erythematous border—is common but not universal. Some lesions may appear uniformly red or have vesicular, crusted, or necrotic centers.